Drone Delivery and Geographic Access to Medications

Research proposal for Planet Education and Research
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Research purpose

I propose to examine when prescription-drone services reduce geographic barriers to medication access, and when those changes translate into medication receipt and continuity of care. The information systems question concerns complementarity: how can digitally coordinated delivery help existing pharmacies and providers reach people facing transport constraints? Benefits are hypotheses to test, not assumed consequences of deployment.

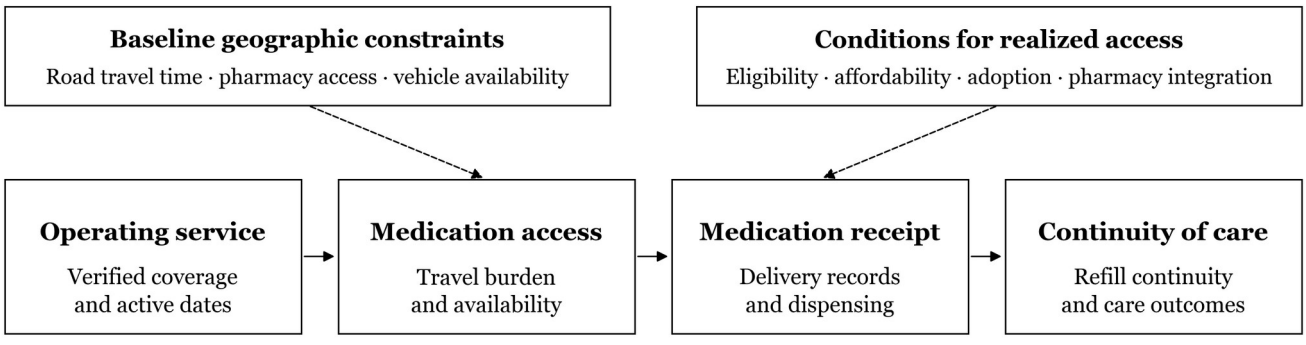
Planet imagery would provide historical evidence of physical sites and their surroundings. I will combine this evidence with verified service histories and road-network accessibility. The design separates operating-service eligibility, realized access, and downstream care outcomes; each requires its own evidence.

Research questions

1. Measurement. Does historical imagery add reproducible information about physical deployment beyond a registry based on regulatory and operational records?

2. Accessibility. Does verified service eligibility reduce medication-access frictions, and how do effects vary with baseline road travel, pharmacy access, and household vehicle availability?

3. Realized benefit. When does availability translate into medication receipt or refill continuity, and who remains excluded by eligibility and adoption barriers?



Hypothesized pathway; downstream outcomes require linked administrative or patient data.

Figure 1. Proposed mechanism from operating-service eligibility to care outcomes. Solid arrows describe hypothesized pathways; dashed arrows indicate contextual conditions. This is a conceptual model, not an empirical result.

Related literature and research gaps

This selected review connects healthcare logistics, patient access, and information systems theory to motivate a focused study of U.S. prescription delivery to patients at home.

What existing research establishes

Healthcare logistics. Haidari et al. [1] use simulation to show how drone vaccine transport can change supply availability and costs under different geographic and operational conditions. Nisingizwe et al. [2] document faster blood-product delivery and reduced wastage in Rwanda. These studies provide evidence about distribution systems; their settings and outcomes do not directly establish benefits for U.S. patients receiving prescriptions at home.

Medication access. Manne et al.'s recent scoping review [3] finds that the medication-delivery literature emphasizes feasibility, delivery performance, regulation, and implementation. Evidence on adherence, clinical outcomes, patient experience, and equity remains limited. Levesque et al. [4] conceptualize access as an interaction between service characteristics and people's abilities to obtain care, supporting a distinction between geographic availability and realized access.

Information systems mechanisms. Strong et al. [5] explain how the possibilities offered by an information system become organizational outcomes through affordance actualization. I use this as a theoretical lens: delivery capability creates a potential benefit that depends on pharmacy integration, eligibility, adoption, and local constraints. The original study concerns EHR implementation; applying the lens to drone delivery is part of this proposal.

Gaps addressed by this study

Gap in the proposed setting	Research response
Measurement of exposure	Reconstruct actual operating periods and historical coverage; record pauses and exits instead of equating announcements or infrastructure with service.
From delivery capability to access	Separate modeled geographic access from verified medication receipt and refill continuity, with evidence requirements for each outcome.
Distribution of benefits	Test whether baseline pharmacy access, travel burden, and vehicle constraints explain heterogeneous effects and persistent exclusion.

Expected contribution

The intended contribution is an auditable measurement approach and a test of the conditions under which autonomous delivery complements healthcare infrastructure. The study will report where inference stops when operational coverage, imagery detectability, outcome access, or the number of independent rollouts is insufficient.

Study setting and the role of Planet imagery

The initial search covers the contiguous United States during 2019–2026, beginning with College Station, Texas, and the Salt Lake Valley, Utah. I will seek approximately 12 months before and up to 24 months after verified starts, subject to archive availability and operating duration. Prescription delivery, general retail delivery, and medical-specimen transport will be coded separately.

A July 2024 report to the Texas pharmacy board records approval on 2 May 2023 and prescription-drone service beginning on 28 August 2023 [7]. Amazon's later public notice states that the College Station pilot ended, without specifying the closure date [8]. Intermountain announced service for selected eligible Salt Lake Valley households on 4 October 2022 [9]. These are documentary starting points; neither implies countywide or continuous coverage.

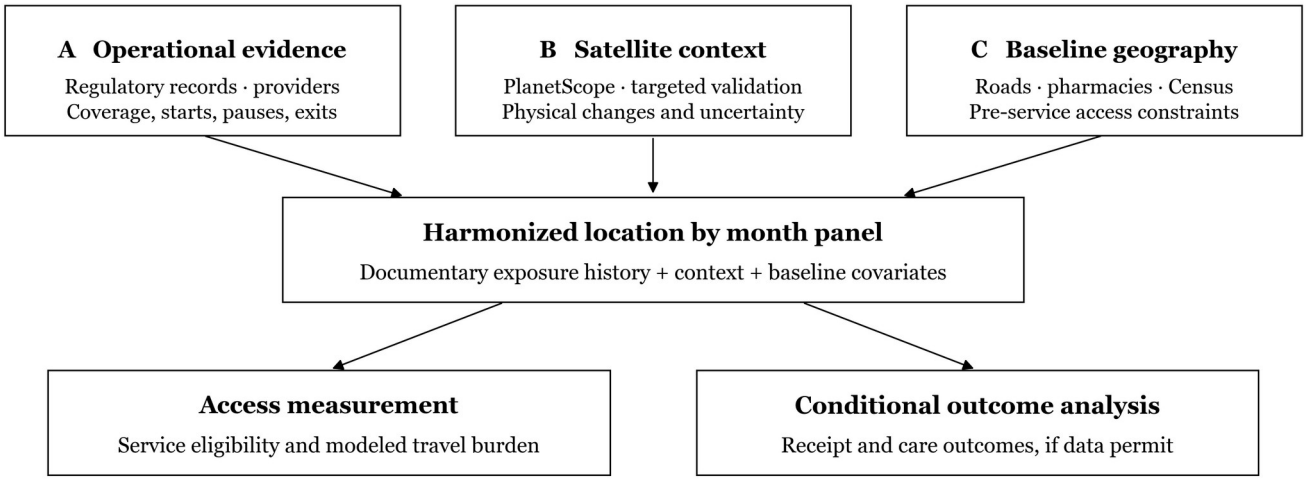


Figure 2. Evidence architecture. Operational records establish service exposure; imagery characterizes physical context. The analysis links these distinct inputs without inferring operating dates, medication receipt, or health effects from imagery alone.

Requested imagery and pilot scope

I request historical PlanetScope orthorectified surface-reflectance imagery with metadata and quality masks. Its approximately 3 m output grid [10] supports a pilot assessment of facility-scale context; it should not be assumed to resolve individual drones, small pads, or delivery activity. If separately authorized, selected 50 cm SkySat imagery [11] could support validation. SkySat and tasking are not included in E&R Basic access [13].

A planning ceiling is 10 candidate sites and 10 comparison sites, each with a 4 km² area of interest and 12 observation dates: 960 km² of area-date requests. This is a sampling calculation, not a guaranteed quota charge. I will confirm archive coverage, download accounting, and current E&R terms [12] before ordering and reduce the pilot if necessary.

Registry and image validation

I will prioritize regulatory and operational evidence, preserve source provenance, and record date, location, and coverage uncertainty. Imagery checks will address cloud, shadow, registration, and seasonality. Two independent annotations and held-out sites or regions will assess agreement and, where labels permit, precision and recall. Last-absent and first-present images bound physical change; they do not establish an exact service start.

Measurement and empirical strategy

Exposure and outcome hierarchy

The primary exposure is eligibility for an operating prescription-delivery service at a location and time. Historical coverage, pauses, and closure dates must be verified. Actual use is a subsequent behavioral choice and will not be substituted for eligibility when estimating service availability effects. Baseline OSM road impedance, pharmacy locations, Census demographics, and vehicle availability will be measured before treatment where possible.

Outcome	Evidence and interpretation
Availability and geographic access	Verified eligibility plus road-network measures. Modeled travel burden is potential access, not observed utilization.
Receipt and delivery performance	Operator or pharmacy records of completed deliveries and time to receipt. These data have not yet been secured.
Continuity of medication supply	Longitudinal dispensing or claims data for refill gaps or refill continuity. Dispensing is not proof of medication ingestion.
Care and health outcomes	Pre-specified endpoints in approved longitudinal patient data, if obtained. Spending or mobility changes alone are not clinical benefits.

Identification and heterogeneity

If the registry yields enough independent rollout units and suitable longitudinal outcomes, I will estimate cohort-specific difference-in-differences and event studies following Callaway and Sant'Anna [6]. The estimand is the effect of service eligibility among eligible locations or patients. Never-treated or not-yet-treated units will be considered as comparisons, conditional on overlap and a defensible parallel-trends assumption.

I will examine pre-treatment trends, selective rollout, concurrent pharmacy or transport changes, seasonality, and spillovers. Inference will be clustered at the rollout decision level; a large number of small geographic cells does not create many independent interventions. Pre-specified heterogeneity will focus on baseline travel burden, pharmacy access, and vehicle availability. Exposure buffers and alternative coverage definitions will probe spillover and classification sensitivity.

Because service can pause or close, an absorbing-treatment estimator is appropriate only within verified uninterrupted operating windows. Exits and interruptions will be analyzed separately. Regulatory approval, weather, and distance will not be treated as automatically valid instruments; each could affect access or outcomes through other pathways.

Limits on interpretation

Outcome access, historical coverage, and the number of independent deployments remain uncertain. With only a few usable sites, I will present a comparative feasibility study rather than a national causal estimate. Without linked outcomes, results will be restricted to service availability and modeled accessibility. No causal or clinical conclusion will be inferred from satellite images alone.

Feasibility and execution plan

The first phase is designed to establish whether the project is measurable and whether a credible outcome study is possible. A larger analysis will proceed only when the registry, imagery pilot, and data agreements support it.

Period	Work and decision
Months 1 and 2	Audit operational sources; reconstruct pilot coverage and active dates; test image detectability. Decide whether imagery adds usable evidence beyond documentary records.
Months 3 and 4	Construct baseline GIS access measures; validate annotations; seek operator or healthcare data agreements and required ethics approval. Define outcomes and comparison units before estimation.
Months 5 and 6	Run the feasible analysis, robustness checks, and heterogeneity tests. Produce a documented registry, measurement protocol, research manuscript, and access maps within license limits.

Fallbacks and decision rules

If relevant infrastructure is not reliably visible, imagery will serve only as environmental context and the registry will remain document-based. If precise service boundaries cannot be verified, uncertain locations will be flagged or excluded rather than assigned broad city or county exposure. If patient or operator data cannot be obtained, the deliverable will be an access-measurement and feasibility study with no claims about receipt, adherence, or health.

Data governance and noncommercial use

This is noncommercial academic research at KAIST, separate from TalentSeeker's business activities. Planet access and U.S. outcome-data access have not yet been secured. Licensed imagery will be used and shared only under applicable terms; public outputs will contain permitted derived material and source documentation. Patient-level work will require appropriate institutional review, data-use agreements, secure processing, and aggregate reporting. No patient records or commercial datasets will be published with the research materials.

Researcher preparation

I am a Ph.D. student in Management Information Systems in KAIST's Business and Technology Management program, advised by Professor Chul Ho Lee. My preparation combines OSM-supported road detection and physics/XGBoost speed-recommendation work, behavioral research on 444 personal-mobility users and non-users using PLS-SEM and multi-group analysis, and a Yuseong District-supported study linking Tashu bike-sharing, transit, spending, and population-movement data.

My healthcare work includes development and study of an LLM/RAG chatbot with Jeonbuk National University Hospital. Alongside doctoral research, I serve as Chief Data Officer at TalentSeeker and teach data science at Baekseok University. These experiences support the project's technical and behavioral components; they do not imply that U.S. operator or patient datasets are already available.

References

Selected scholarly literature and primary operational and product sources. Web sources checked 30 October 2026.

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[3] Manne, N., Coustasse, A., Kimble, C., Simonton, D., & Pile, T. (2026). Drone-supported medication delivery as a rural pharmacy access strategy: A scoping review of logistics, regulation, and pharmacist integration. Journal of the American Pharmacists Association, advance online publication, 103553. doi: 10.1016/j.japh.2026.103553

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[5] Strong, D. M., et al. (2014). A Theory of Organization-EHR Affordance Actualization. Journal of the Association for Information Systems, 15(2), 53–85. doi: 10.17705/1jais.00353

[6] Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-Differences with multiple time periods. Journal of Econometrics, 225(2), 200–230. doi: 10.1016/j.jeconom.2020.12.001

[7] Amazon Pharmacy (2024, July 25). Prescription drone-delivery pilot report submitted to the Texas State Board of Pharmacy, August 2024 board materials, item D.1. Source

[8] Amazon. Amazon Pharmacy prescription drone-delivery announcement and subsequent College Station closure notice. Source

[9] Intermountain Healthcare (2022, October 4). Announcement of Zipline and Intermountain drone deliveries beginning in the Salt Lake Valley. Source

[10] Planet. PlanetScope imagery documentation. Source

[11] Planet. SkySat imagery documentation. Source

[12] Planet. Education and Research Program. Source

[13] Planet Support. SkySat data and Tasking Dashboard access in the Education and Research Basic Program. Source